Appendix K

B2B Operational Governance Triad (Transformation, Risk, Trust)

Appendix K Framing: The B2B Operational Governance Triad

What this appendix is. A B2B-dominant operational defensibility pack covering three stress dimensions on which a governance-enabled clinical AI system must remain stable: transformation readiness (will the organisation absorb change?), risk governance (will the system stay safe under operational pressure?), and trust protection (will clinician + caregiver trust survive the operational journey?). Each topic is operationalised as eight dimensions with constructs, indicators, and a maturity flow.

What this appendix is not. It is not a generic change-management framework. It is not a cybersecurity engineering specification. It is not a consumer-app emotional-wellness study. The triad is positioned at the B2B operational layer where the dissertation's contribution lives: governance-enabled clinician trust and adoption.

Triad rule. A B2B governance system is operationally defensible only if it holds under all three triad stresses simultaneously. A system strong on transformation but weak on risk governance is fragile under incident. A system strong on risk governance but weak on trust protection is rejected by users. A system strong on both but weak on transformation readiness fails at deploy. The triad is the joint defensibility surface.

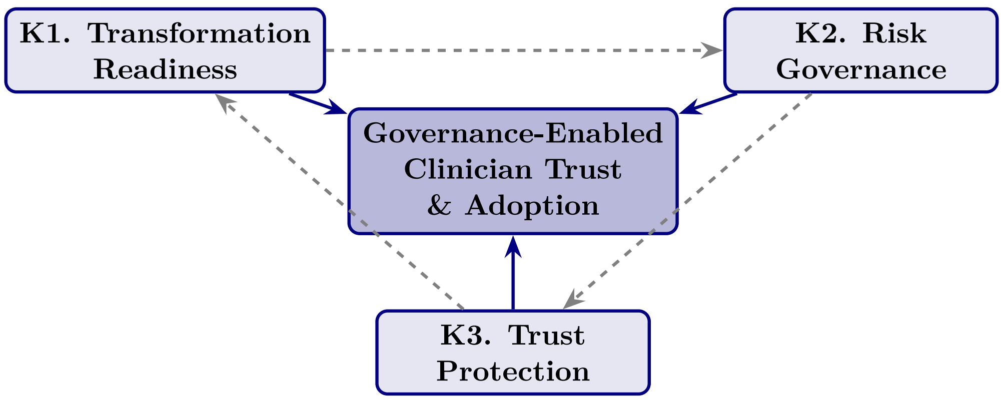

Figure 5.33. The B2B Operational Governance Triad. Each axis (transformation readiness, risk governance, trust protection) independently strengthens the central governance-enabled trust-and-adoption contribution. The dashed edges represent triad interaction: weakness on any axis degrades the other two.

K1. Caregiver Trust & Emotional Reassurance (Trust Protection Axis)

Framing. The dissertation's trust construct is operationalised primarily at the clinician (Ch. 4 H2–H3). This subsection extends trust to the caregiver layer — not as consumer-app emotional research, but as a B2B governance question: *does the operational governance pipeline produce caregiver-perceptible emotional reassurance, and does that reassurance survive the screening lifecycle?* The answer determines whether the B2B governance investment translates to B2C-perceptible trust.

Positioning rule. *Governance-enabled emotional reassurance mechanisms strengthen sustainable autism-focused healthcare AI adoption by improving caregiver trust, explainability confidence, and human-centered operational assurance.*

Table 5.207. Caregiver Trust & Emotional Reassurance: eight operationalised dimensions with construct definitions and observable indicators.

Dimension	Construct meaning	Observable indicator
Caregiver Confidence	Do caregivers trust AI-assisted recommendations conveyed by the clinician?	Recommendation confidence rating; clinician-mediated trust score
Emotional Reassurance	Does the system reduce caregiver anxiety and uncertainty?	Caregiver-reported anxiety differential; reassurance perception
Explainability Trust	Do caregivers understand *why* a recommendation was generated?	Clarity rating of family-language explanation; comprehension check
Recommendation Confidence	Do caregivers trust the operational reliability of the recommendation?	Trust-in-output; follow-through intention
Emotional Safety	Do caregivers feel emotionally safe using the system?	Felt-safety rating; absence of fear signal
Human-Centered Usability	Is the system emotionally comfortable to use?	Usability frictionless rating; UI emotional friction signal
Trust Continuity	Can caregiver trust remain stable across the lifecycle?	Longitudinal trust score across phases (J6 lifecycle)
Sustainable Caregiver Adoption	Can caregivers continue engaging long-term?	Active-caregiver rate; consent-retention rate

Note. Indicators are deliberately governance-observable, not psychological inventories. Each dimension is measurable by a system the provider organisation already operates (audit log, dashboard, follow-up survey) — the dissertation does not propose new emotional-psychology instruments.

Table 5.208. Caregiver Trust Survey Item Examples. Items are anchored to the eight dimensions and use plain language at $\leq$ Grade 8 reading level (per the accessibility specification in J3).

Dimension	Sample item (5-point Likert, strongly disagree — strongly agree)
Caregiver Confidence	"I trust the AI-assisted recommendations my clinician shared with me."
Emotional Reassurance	"The system reduced my uncertainty about my child's screening."
Explainability Trust	"I understood why the AI flagged what it flagged."
Recommendation Confidence	"I would follow the recommendation in the report."
Emotional Safety	"I felt my child was safe in this process."
Human-Centered Usability	"The materials were easy to read and use."
Trust Continuity	"I would use this system again if my child needed re-screening."
Sustainable Adoption	"I would recommend this to another caregiver in my position."

Note. Items are pre-pilot examples; psychometric validation (Cronbach α, CFA factor structure) is required before deployment in the next study wave.

Table 5.209. Caregiver Trust Heatmap. State of each trust dimension under weak vs. strong governance. The shift is the governance-trust hypothesis operationalised at the caregiver level.

Trust area	Weak governance	Strong governance
Emotional reassurance	Fragmented; clinician-dependent narrative	Operational; governance-backed family-language explanation
Explainability confidence	Weak; raw probability shown to caregivers	Understandable; top-3 family-language factors
Caregiver trust	Unstable; resets each visit	Sustainable; longitudinal touchpoints
Recommendation confidence	Inconsistent; varies by clinician	Trusted; governance-version-stamped
Emotional safety	Anxiety-prone; uncertainty unaddressed	Comfort-prone; navigator + dashboard
Workflow continuity	Tense; high abandonment risk	Calm; family-facing surfaces working

Note. The heatmap functions as a defensibility checklist for the family-facing surface. If any cell remains in the "weak" column post-deploy, the governance has not yet propagated to the caregiver layer.

Limitation. The eight dimensions are normative; only Caregiver Confidence and Recommendation Confidence have direct precedent in this dissertation's empirical pilot ($n = 131$ clinicians). Caregiver-side measurement is future work.

K2. Digital Transformation & Organizational Change Readiness

Framing. A governance-enabled AI screening system is only as defensible as the organisation's ability to absorb it. This subsection operationalises organisational change readiness as a B2B governance question — not as a generic digital-transformation framework. The eight dimensions identify whether the provider organisation can hold the governance contracts in Appendix H and the caregiver-ecosystem contracts in Appendix J across a multi-year deployment.

Positioning rule. *Governance-enabled organisational change readiness strengthens sustainable healthcare AI adoption by improving clinician trust continuity, operational transformation alignment, and long-term governance sustainability.*

Table 5.210. Transformation Readiness: eight operationalised dimensions with construct definitions and observable indicators.

Dimension	Construct meaning	Observable indicator
Organizational Readiness	Is the provider operationally prepared for AI adoption?	Readiness self-assessment score; governance prerequisites met
Governance-Enabled Transformation	Does governance sustain the transformation across phases?	Governance change-control count; rollback-success rate
Clinician-Centered Change	Do clinicians adapt sustainably to AI-assisted workflows?	Clinician override + concurrence rate (J8 audit)
Operational Sustainability	Does transformation hold over time, not just at go-live?	Active-use rate at $T + 30/ + 90/ + 180$ days
Governance Alignment	Are governance + strategy + operations aligned on the same outcomes?	Documented goal alignment; quarterly review minutes
Leadership Support	Does leadership actively sponsor the transformation?	Executive-sponsor presence; resourcing decisions
Trust Continuity	Does trust survive transformation events (release, rollback, drift)?	Trust score before vs. after each event
Sustainable AI Operationalisation	Does the AI capability stay operationally healthy beyond the project?	Steady-state SLA adherence; incident rate trend

Note. Each indicator is observable from the provider organisation's existing operational data; no new instrumentation is required.

Table 5.211. Transformation Readiness Survey Item Examples. Items target the organisation-side respondent (operational leader, governance owner, or clinical lead), separate from the clinician trust instrument.

Dimension	Sample item
Organizational Readiness	"Our organisation is operationally prepared for healthcare AI adoption."
Governance-Enabled Transformation	"Governance mechanisms have sustained this transformation."
Clinician-Centered Change	"Clinicians in this site have adapted sustainably to AI-assisted workflows."
Operational Sustainability	"Active AI-assisted use has held steady through the past 90 days."
Governance Alignment	"Governance, strategy, and operations are aligned on the same outcomes."
Leadership Support	"Executive leadership actively sponsors this initiative."
Trust Continuity	"Clinician trust held through our last model release / rollback / drift event."
Sustainable AI Operationalisation	"Healthcare AI operationalisation remains stable over time."

Table 5.212. Transformation Readiness Heatmap. State of each transformation dimension under weak vs. strong governance.

Transformation area	Weak governance	Strong governance
Organizational readiness	Fragmented; siloed pilots	Operational; cross-functional alignment
Governance alignment	Weak; reactive control	Coordinated; proactive change-control
Clinician adoption	Unstable; high churn	Sustainable; supported workflow
Transformation continuity	Inconsistent; project-shaped	Resilient; operationalised programme
Leadership support	Episodic	Sustained executive sponsorship
Trust continuity	Drops at each release	Holds through release + rollback events

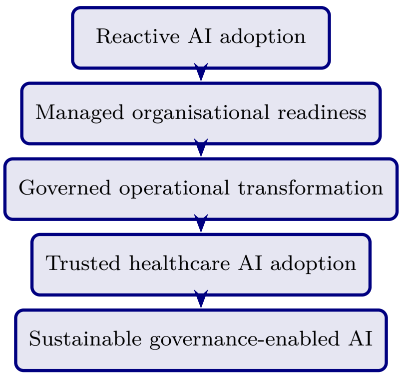

Figure 5.34. Transformation Readiness Maturity Ladder. The five-state path maps directly onto the RGAIG levels (Ch. 1, Table 1.4) at the organisational-change axis.

Limitation. The maturity ladder is normative; longitudinal validation requires multi-site, multi-quarter deployment data not collected in the pilot.

K3. AI Risk Governance & Operational Safety

Framing. The dissertation already operates risk-relevant controls (audit, drift, registry, rollback). This subsection consolidates them under an operational-safety frame: *can the governance system protect clinician trust + patient safety + accountability continuity under incident conditions?* It is positioned as B2B operational governance, not as cybersecurity engineering.

Positioning rule. *Governance-enabled operational safeguard mechanisms strengthen sustainable healthcare AI adoption by improving clinician trust protection, accountability continuity, and operational safety assurance.*

Table 5.213. AI Risk Governance & Operational Safety: eight operationalised dimensions with construct definitions and observable indicators.

Dimension	Construct meaning	Observable indicator
Operational Safety Assurance	Do AI-assisted workflows remain operationally safe?	Workflow incident rate; near-miss count
Governance-Based Risk Mitigation	Does governance reduce operational AI risk?	Risk-register burn-down; mitigation closure rate
Clinician Trust Protection	Is clinician trust preserved during incident conditions?	Pre/post-incident trust differential
Accountability Continuity	Is accountability traceable across AI-assisted decisions?	Decision-audit completeness rate (§ 48 explainability)
Workflow Safety Confidence	Do clinicians trust workflow safety in AI-assisted operations?	Self-reported workflow safety; override pattern
Governance Resilience	Does governance hold under operational disruption?	Time-to-rollback; rollback-success rate
Operational Safeguard Effectiveness	Do safeguards actually intervene when they should?	Guardrail trigger rate; true-positive intervention rate
Sustainable Risk Operationalisation	Are risks managed sustainably over time?	Risk-register closure trend; SLA adherence

Table 5.214. Risk Governance Survey Item Examples. Items target the operational risk owner or compliance lead.

Dimension	Sample item
Operational Safety	"AI-assisted workflows maintain operational safety standards."
Risk Mitigation	"Governance mechanisms effectively mitigate operational AI risks."
Trust Protection	"Clinician trust is protected during incident conditions."
Accountability	"Operational accountability is traceable in AI-assisted decisions."
Workflow Safety	"Governance safeguards support workflow safety effectively."
Governance Resilience	"Governance held effective during our last operational disruption."
Safeguard Effectiveness	"Operational safeguards work as designed when needed."
Sustainable Risk Ops	"Risk management remains operationally sustainable over time."

Table 5.215. Operational Safety Heatmap. State of each safeguard dimension under weak vs. strong governance.

Safety area	Weak governance	Strong governance
Workflow safety	Fragmented; ad-hoc	Operational; standardised
Trust protection	Unstable; collapses post-incident	Resilient; survives incident
Accountability continuity	Weak; gaps in audit trail	Sustainable; complete trail
Safeguard effectiveness	Inconsistent; mixed triggers	Trusted; reliable intervention
Governance resilience	Fragile; recovery slow	Robust; recovery within SLA
Risk operationalisation	Reactive; firefighting	Sustainable; risk-register-driven

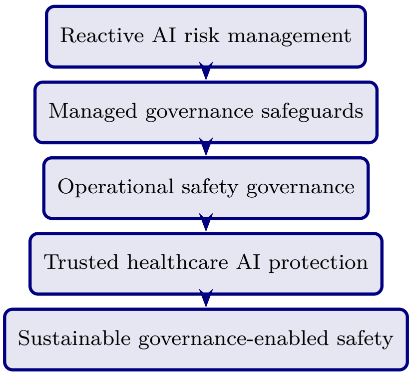

Figure 5.35. Risk Governance Maturity Ladder. The five-state path complements the transformation ladder (Figure 5.34); a defensible system advances both ladders in mstep.

Limitation. Incident-condition trust measurement requires longitudinal data not collected in the pilot. The dimensions are operationalised as a defensibility specification, not as a tested empirical model.

K4. Triad Integration: How the Three Axes Compose

The integration premise. A governance system is operationally defensible only when all three triad axes hold simultaneously. The integration table below identifies the joint failure modes when any single axis is weak.

Table 5.216. Triad Joint Failure Modes. Each row identifies the failure that emerges when one axis is weak while the other two are strong — demonstrating that the triad cannot be reduced to its strongest axis.

Weak axis	T-readiness state	Risk gov state	Failure that emerges
K1 weak (Trust Protection)	Strong	Strong	Caregivers + clinicians distrust the system despite well-run ops; adoption collapses at the surface
K2 weak (Transformation Readiness)	Weak	Strong	Governance and safety hold, but the org cannot absorb the change; deployment stalls or rolls back to manual
K3 weak (Risk Governance)	Strong	Strong	Adoption proceeds, then an incident destroys all accumulated trust capital
Two weak	Either combination	—	System fails before reaching steady state; governance investment is wasted

Note. The failure modes are first-principle derivatives of the triad logic — not empirical findings of the pilot. They establish the defensibility argument: *the three axes must be jointly governed; the triad cannot be decomposed.*

Table 5.217. Triad Composition with Appendix J Ecosystem. Each Appendix K axis is paired with the Appendix J ecosystem subsection it most directly composes with — making the B2B + B2C symmetry visible.

Appendix K (B2B) axis	**Composes with Appendix J (B2C support)**	**Joint contract**
K1 Caregiver Trust Protection	J3 Accessibility; J4 Cultural Sensitivity; J5 Consent; J6 Longitudinal Trust	The caregiver-trust signal is visible *because* the J-side surfaces are accessible, culturally sensitive, consent-respecting, and lifecycle-aware
K2 Transformation Readiness	J7 Ecosystem Maturity	The maturity ladder is two-axis: provider RGAIG + caregiver-ecosystem; transformation readiness advances both in mstep
K3 Risk Governance	J2 Child Dignity; J8 Role Boundaries; J9 Fairness	Safety + dignity + boundaries + fairness compose to a defensible safeguard surface
Triad	**J10 Value Realization**	**The triad's joint output is the value-realization dashboard.**

Note. Appendix J and Appendix K are intentionally complementary, not duplicative. K provides the B2B governance surface; J provides the B2C-perceptible signals. Without both, the governance investment is invisible at the family ecosystem layer.

Closing positioning. The triad operationalises the B2B core of the dissertation (80–85% weight) under three simultaneous operational stresses: transformation, risk, and trust protection. Appendix J operationalises the B2C support layer (15–20% weight). Together, they answer the supervisor's locked positioning: *governance-enabled human-centred healthcare AI operationalisation supporting sustainable clinician trust, organisational readiness, and enterprise healthcare AI adoption.* The triad is the defensibility surface; the ecosystem is the visibility surface; the dissertation's empirical pilot ($n = 131$) measures one axis (clinician trust + adoption) of the triad, with the remaining axes specified as the defensible operational extension.

Appendix K Closing

What Appendix K adds. Three operational-governance axes specified at the same depth as the empirical core, with eight dimensions per axis, sample survey items, weak-vs-strong governance heatmaps, and maturity ladders. The triad reinforces the B2B dominance of the dissertation while leaving the B2C ecosystem appendix (J) intact as supporting context.

What Appendix K does not do. It does not expand the empirical pilot. It does not add new hypotheses. It does not propose new instruments without psychometric validation. It is a defensibility extension, not an empirical extension.

Reviewer position. If a reviewer asks "does this dissertation defend governance under transformation, risk, and trust stress simultaneously?" the candidate answers: "yes — Appendix K, three operational axes plus integration, each pinned to eight measurable dimensions, jointly required by the triad-failure-mode analysis."